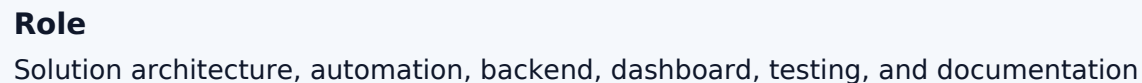


AI-assisted lead operations and CRM automation

A production-style portfolio project for the fictional NorthStar Home Services. The system captures leads, applies deterministic business rules with optional AI assistance, routes qualified opportunities, synchronizes HubSpot, triggers actions, and makes every failure visible.



The problem and the solution

A business workflow designed around speed, control, and visibility

The operational problem

Service businesses receive leads through multiple channels. Without one canonical workflow, requests can be duplicated, delayed, misrouted, or silently lost.

The engineering response

LeadFlow AI persists the request first, returns a fast receipt, and continues the heavier work asynchronously through a controlled orchestration pipeline.

The safety model

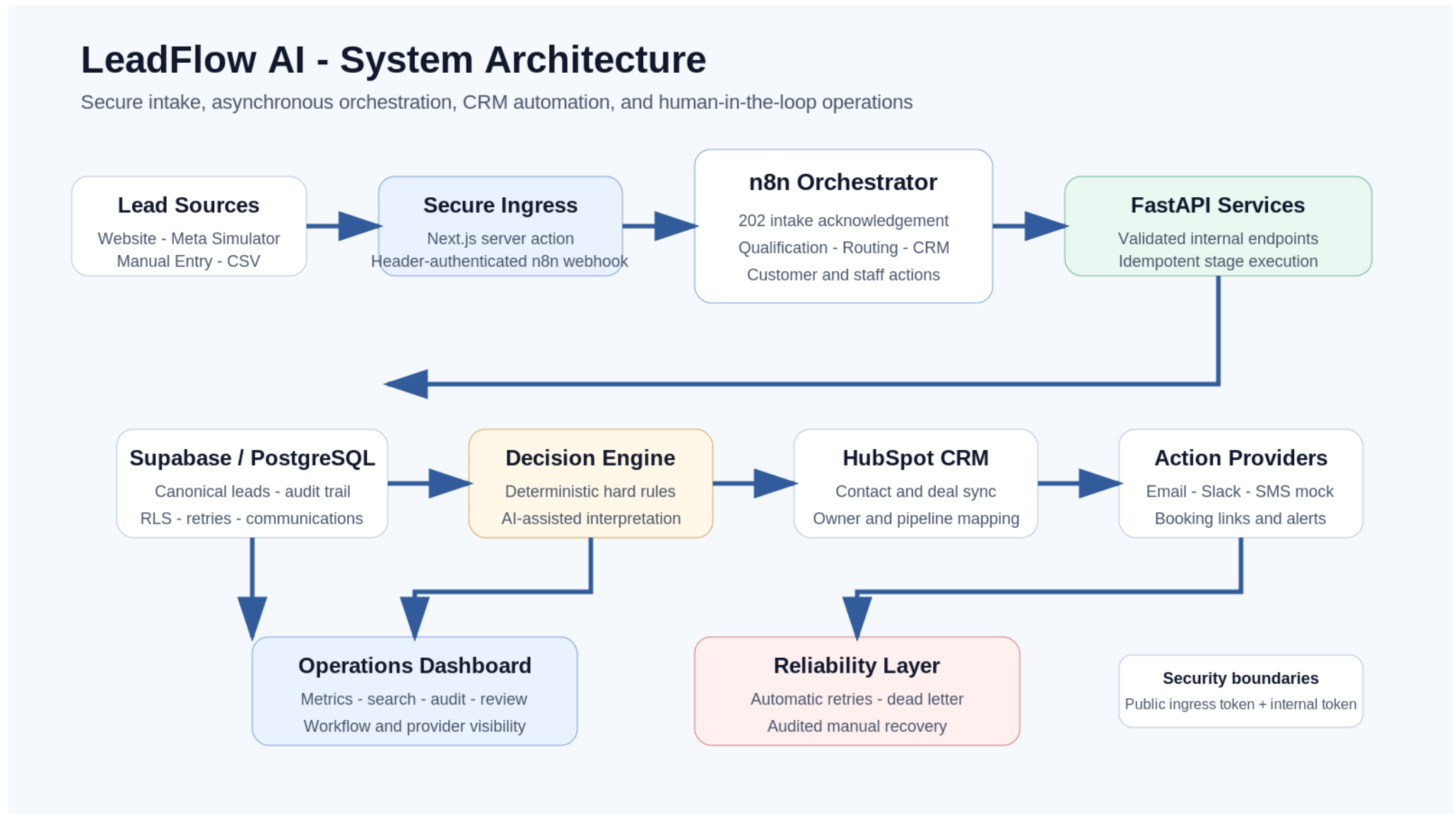
Deterministic hard rules remain authoritative. AI can assist interpretation, but low-confidence or conflicting cases move to human review.

Core outcomes demonstrated

- One canonical lead record across sources
- Authenticated public and internal service boundaries
- HTTP 202 acknowledgement with traceable identifiers
- Replay-safe CRM and communication side effects
- Human review instead of forced automation
- Automatic retries and dead-letter recovery
- Operator visibility through dashboard and audit history

System architecture

Public intake and internal orchestration use separate security boundaries



Secure intake and asynchronous acknowledgement

The lead is persisted before the customer receives confirmation

The screenshot shows the 'Request service' form on the NorthStar Home Services website. The form is titled 'Request service' and includes the following fields: Full name, Email, Phone, Service (dropdown), Location, Urgency (dropdown), Preferred contact (dropdown), and What do you need help with? (text area). The form is set against a light blue background with the NorthStar Home Services logo and a 'Staff sign in' link in the top right corner.

The screenshot shows the 'Request received' confirmation page on the NorthStar Home Services website. The page features a green box with the text 'Request received' and 'We have your service request.' Below this, it states 'The appropriate NorthStar team will review it and follow up.' A white box displays the 'REFERENCE ID' as 'corr_b7b301789afe487a81ed7890528442a8' and includes the instruction 'Keep this reference in case you need to follow up about your request.' The page also includes the NorthStar Home Services logo and a 'Staff sign in' link in the top right corner.

1. Public intake

The public form collects structured lead data. Secret handling stays server-side inside Next.js.

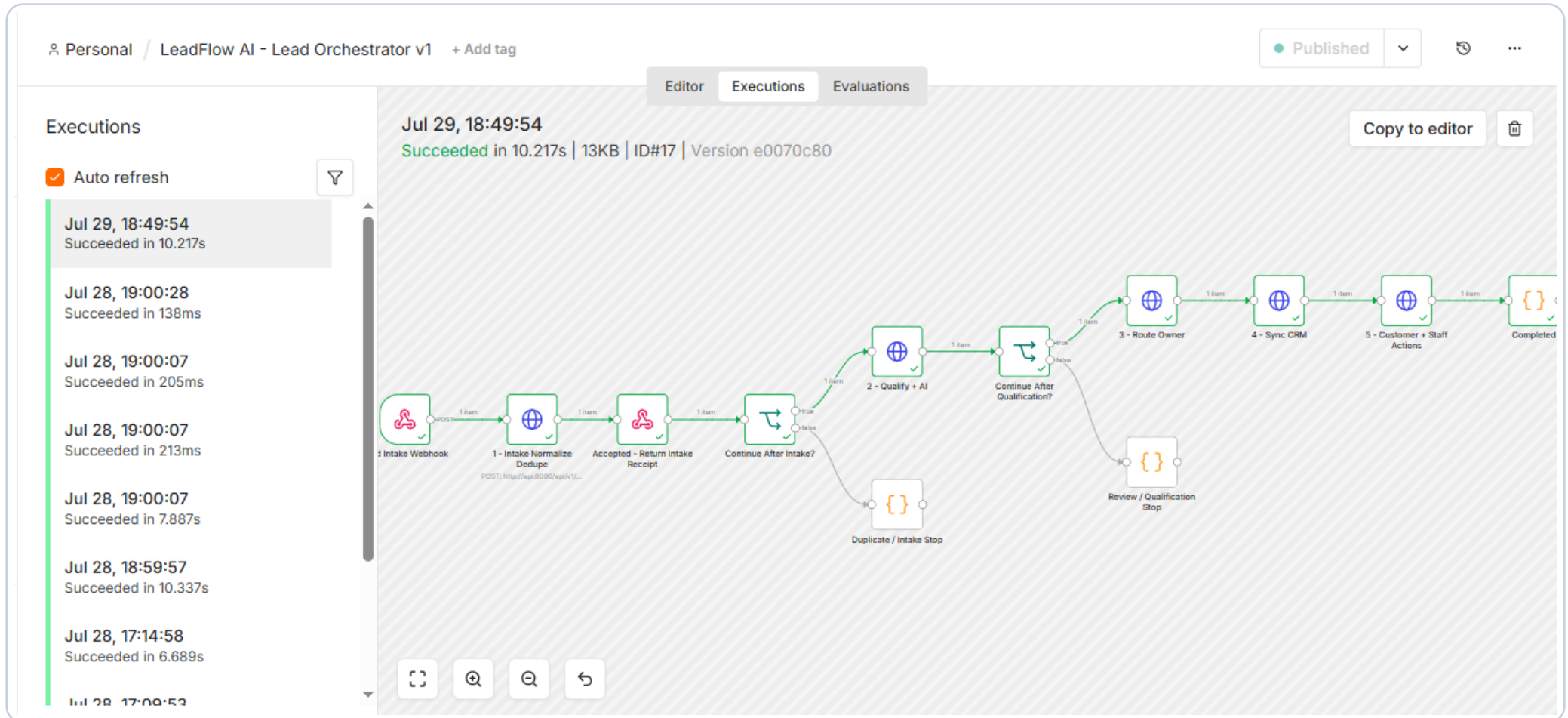
2. Traceable receipt

The customer receives a full reference ID after the intake stage returns HTTP 202.

Security design: the browser does not receive the n8n ingress token. The public webhook validates one token, while n8n uses a separate internal orchestrator token when calling FastAPI. The API validates, normalizes, deduplicates, and stores the lead before acknowledgement.

Workflow orchestration and controlled decisions

n8n coordinates the stages while FastAPI owns the business logic



Fast response

The receipt is returned after intake, not after the full workflow.

Branch safety

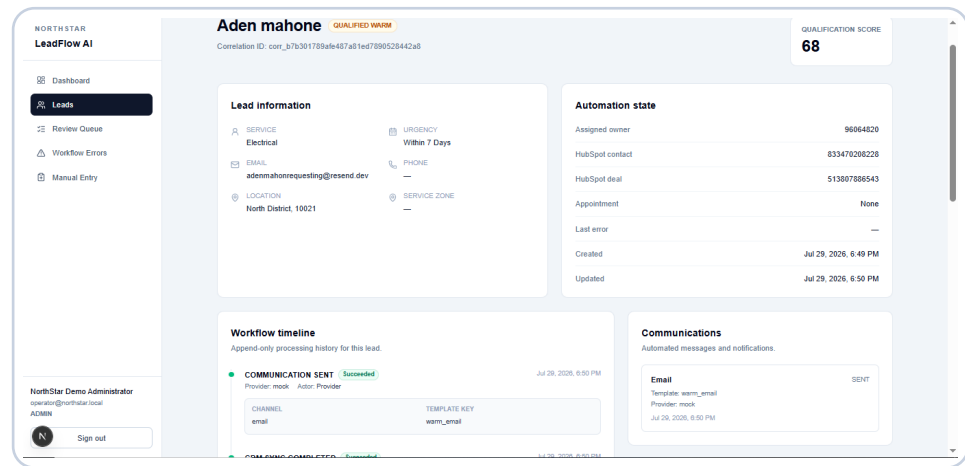
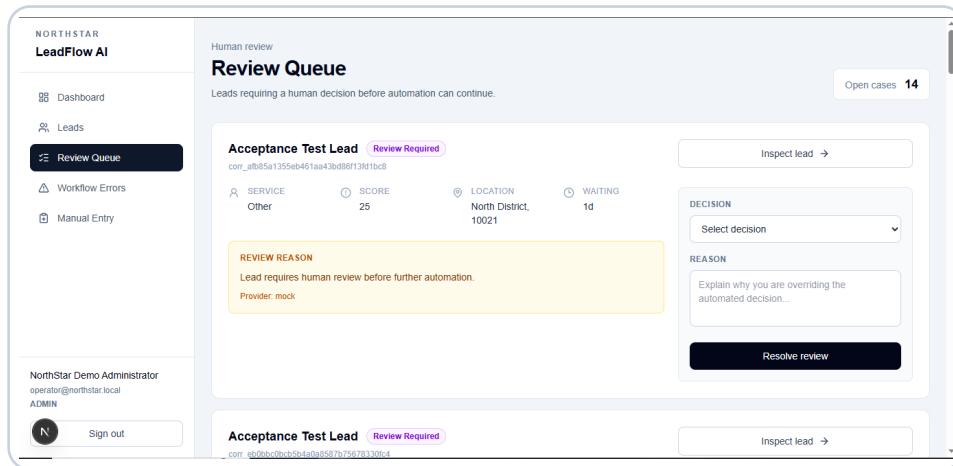
Duplicates stop safely; ambiguous leads enter human review.

Replay safety

CRM and actions preserve completed side effects across retries.

Human-in-the-loop operations

Ambiguous cases are reviewable, explainable, and auditable



Review queue

Operators inspect the reason, select a decision, and record a justification before automation continues.

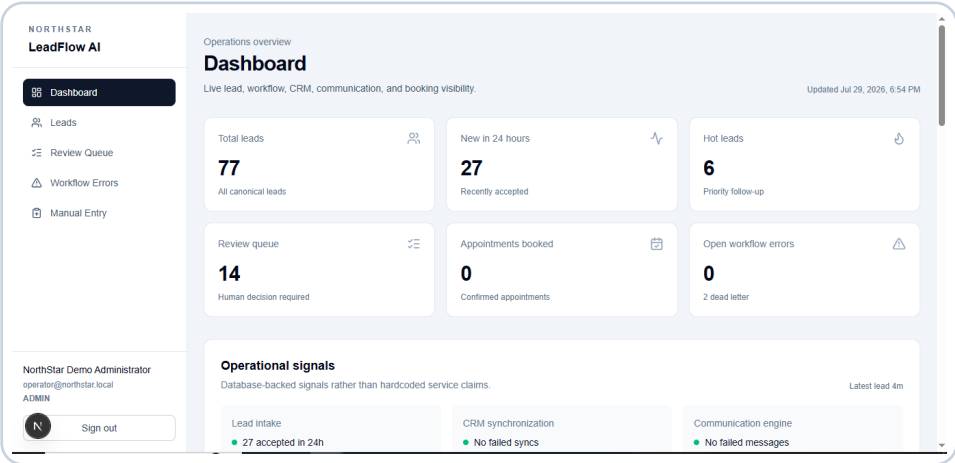
Lead audit workspace

Each lead exposes qualification, routing, CRM, communication, appointment, and workflow-event history.

Design principle: AI is advisory. Unsupported services, out-of-area leads, and other deterministic rules remain authoritative. Low-confidence or conflicting interpretation becomes REVIEW_REQUIRED rather than an unbounded automated decision.

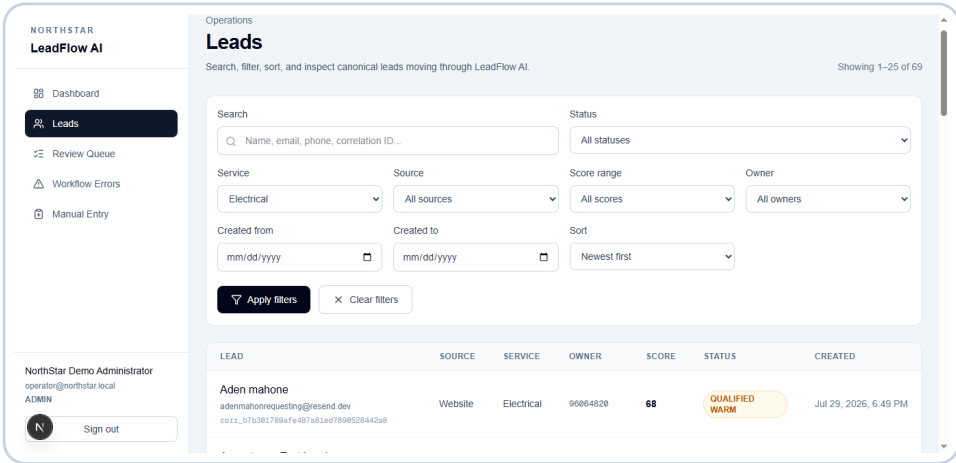
Operational dashboard and lead workspace

Database-backed visibility for daily operations



Executive dashboard

Lead volume, recent intake, hot leads, review demand, bookings, errors, and operational signals.



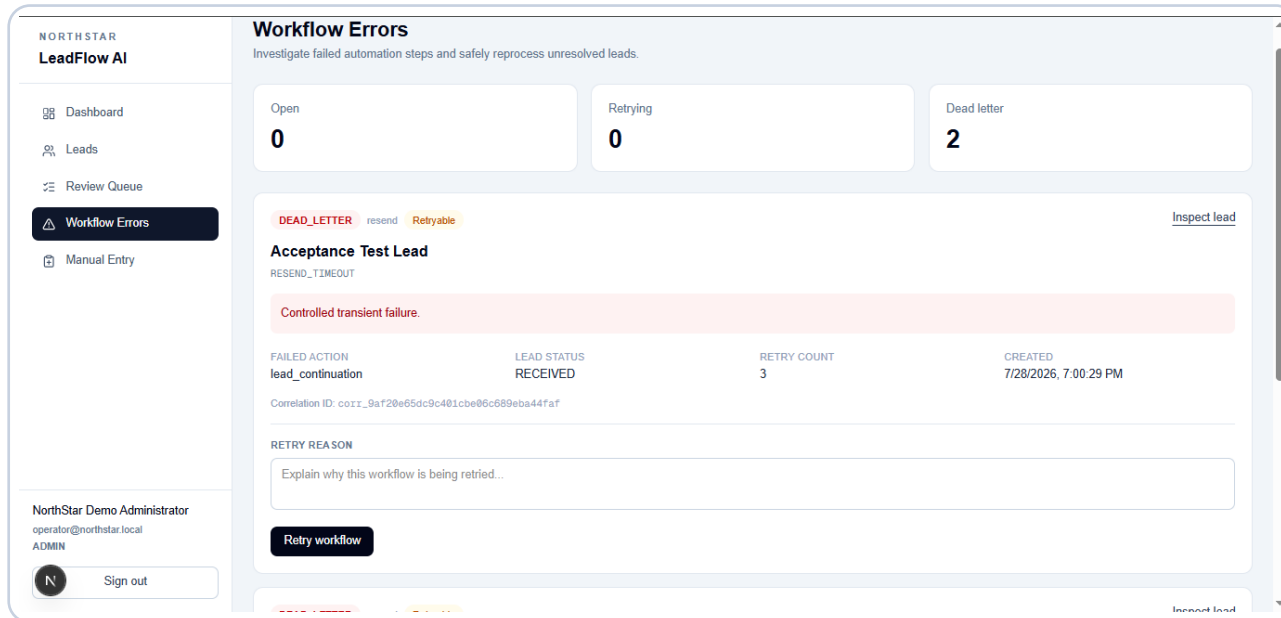
Lead workspace

Search, filters, sorting, exact counts, pagination, and drill-down into the canonical lead record.

Operators can search by customer details or correlation ID and filter by status, service, source, score, owner, and date. The dashboard uses database-backed metrics rather than hardcoded service claims.

Reliability and recovery

Failures remain visible, retryable, and auditable



Automatic retry

Retryable provider failures use controlled exponential backoff and a maximum-attempt policy.

Dead letter

Exhausted or non-retryable failures remain visible and generate a staff alert.

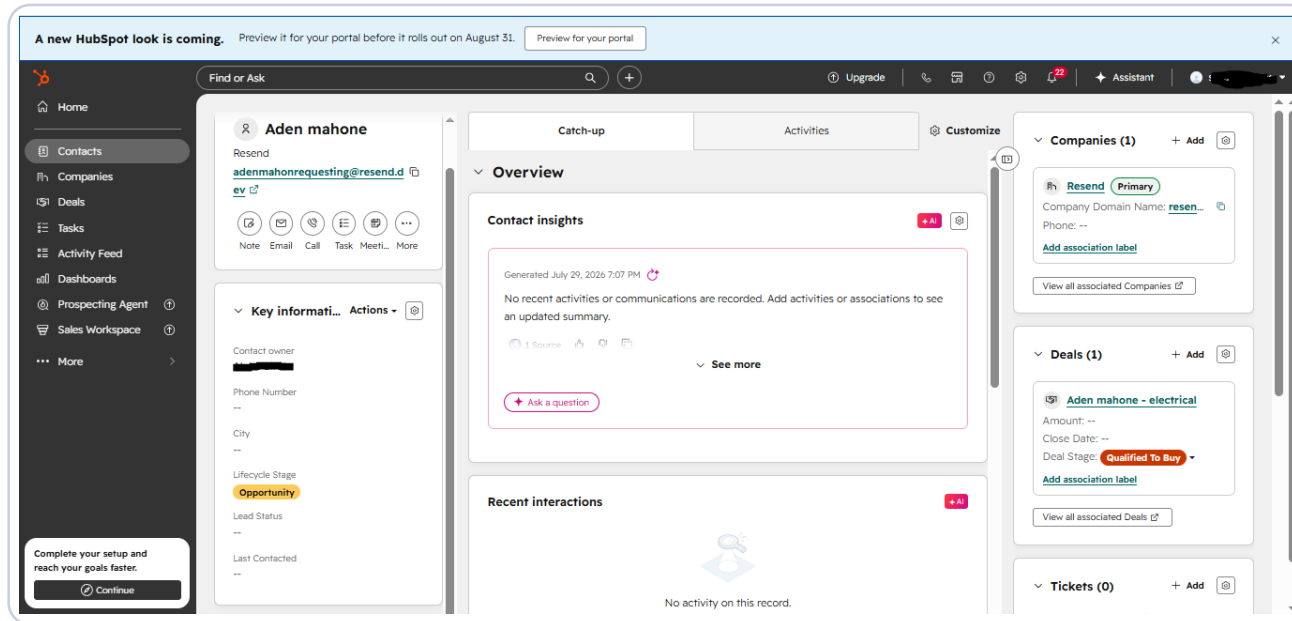
Manual recovery

Administrators can retry safely without duplicating successful CRM or communication actions.

Reliability state changes are persisted as workflow events. This preserves the principle that no lead or provider failure should disappear silently.

HubSpot integration and validation

Provider synchronization is designed for replay safety



CRM synchronization

Qualified leads create or update HubSpot contacts and deals with owner and pipeline mapping.

Provider identity

Stored provider IDs allow repeated CRM stages to preserve the same records.

Validation

The completed local build passed the security, workflow, replay, review, retry, and dead-letter acceptance suite.

Testing and evidence use synthetic identities and a fictional company. No client revenue, conversion improvement, or live production outcome is claimed.

Project closeout

What this portfolio project proves

Architecture

I can design a multi-service automation system with explicit public and internal trust boundaries.

Business logic

I can translate service rules into deterministic, configurable qualification and routing behavior.

AI control

I can use AI as a constrained assistant rather than an unchecked source of truth.

Integrations

I can connect web intake, n8n, APIs, PostgreSQL, HubSpot, and communication providers.

Operations

I can build review, audit, dashboard, search, retry, and recovery workflows for real operators.

Delivery discipline

I can validate the system through repeatable Docker setup and acceptance tests.

LeadFlow AI is now complete as portfolio project #1. The strongest next step is to present the workflow clearly, publish the repository, record the short demo, and use the case study as evidence when applying for AI automation, internal-tool, CRM integration, and business-process automation work.